Layer	Selected baseline	Responsibility
PC client	Ubuntu 24.04 / Python 3.12 PySide6 6.8.3 / GStreamer 1.24	Player, AEC, sensor controls, disk ring
Clip processing	FFmpeg 6.1 / local SQLite index	Assemble and review audience clips
API / storage	FastAPI 0.162.0 / PostgreSQL 16 Redis 7.2	Sessions, queue, event records
Inference worker	Python 3.11 / PyTorch 2.9.1 PANNs + MediaPipe 0.10.21	GPU audio, CPU facial analysis, fusion
Deployment	EC2 g4dn.xlarge / Docker Compose Kubernetes-compatible images	Initial host and portable server package